Discuss the modern DBMS. Talk about how it evolved, why its important and 4 of its key features.

DBMS(Database Management System):

Evolution:

Eerily Days:

In the early days, data was stored in flat files(just like CSV file, comma separated values file) or hierarchical database, which requires significant manual effort for data retrieval. As business grew and data volumes surged, a need for more efficient ways to store and access data emerged.

RDBMS:

Because there was a need for a more efficient and effective to access data, RDBMS was developed in 1970s, it revolutionized the data storage and retrieval by representing data in tables, allowing for more versatile and efficient querying.

NoSQL database:

The turn of the century brought with it the challenged of handling the Internet's explosion and the vast amounts of unstructured data it generated. Modern database system began to diversify, incorporating NoSQL database to cater to needs that traditional RDBMS couldn't address.

Why is DBMS important:

What is a database? It's a shared collection of related data used to support the activities of a particular organization. It can be viewed as a repository of data that is defined and accessed by various users. Different users of the database will have different privileges, everybody can access the data they are allowed to access. The data that they are allowed to access is directly related to their job responsibilities.

For instance, if financial department can't access the financial data stored in database, it won't be able to read and write the financial data into the database, so financial department won't be able to deliver the financial reports for the current month in time, which will cause significant effect to the business.

In a nut shell, modern organizations like companies, government, schools need a reliable way to store data, and DBMS is the reliable way to store data.

Four Key features:

- Data Security: DBMS provides robust security features to protect data from unauthorized access. This includes user authentication, access control, and encryption to ensure data confidentiality.
- Data Integrity: Ensuring data accuracy and consistency is crucial. DBMS enforces data integrity rules to maintain the correctness of data across the database.
- Concurrency Control: DBMS supports concurrent access by multiple users, allowing them to interact with the database simultaneously without compromising data integrity.
- Backup and Recovery: DBMS includes mechanisms for data backup and recovery to protect against data loss. This ensures that data can be restored to a consistent state in case of failures.

Modern Database Management Explained (five.co)
Types of Modern Databases: Overview and Definitions | Zuar
Key Features of DBMS (Database Management Systems) (almabetter.com)